

SS-2005-4 | 塑性崩塌機制：剪力圖 (SFD) 與彎矩圖 (BMD)

Propped Cantilever, T型梁, $L = 600 \text{ cm}$, 均布載重 $wu = 0.0658 \text{ tf/cm}$

T 型斷面

翼板：40 × 2.6 cm
腹板：24 × 2.4 cm
 $Z = 813 \text{ cm}^3$

塑性彎矩 M_p

$$\begin{aligned} M_p &= F_y \times Z \\ &= 2.5 \times 813 \\ &= \mathbf{2032.5 \text{ tf}\cdot\text{cm}} \end{aligned}$$

崩塌載重

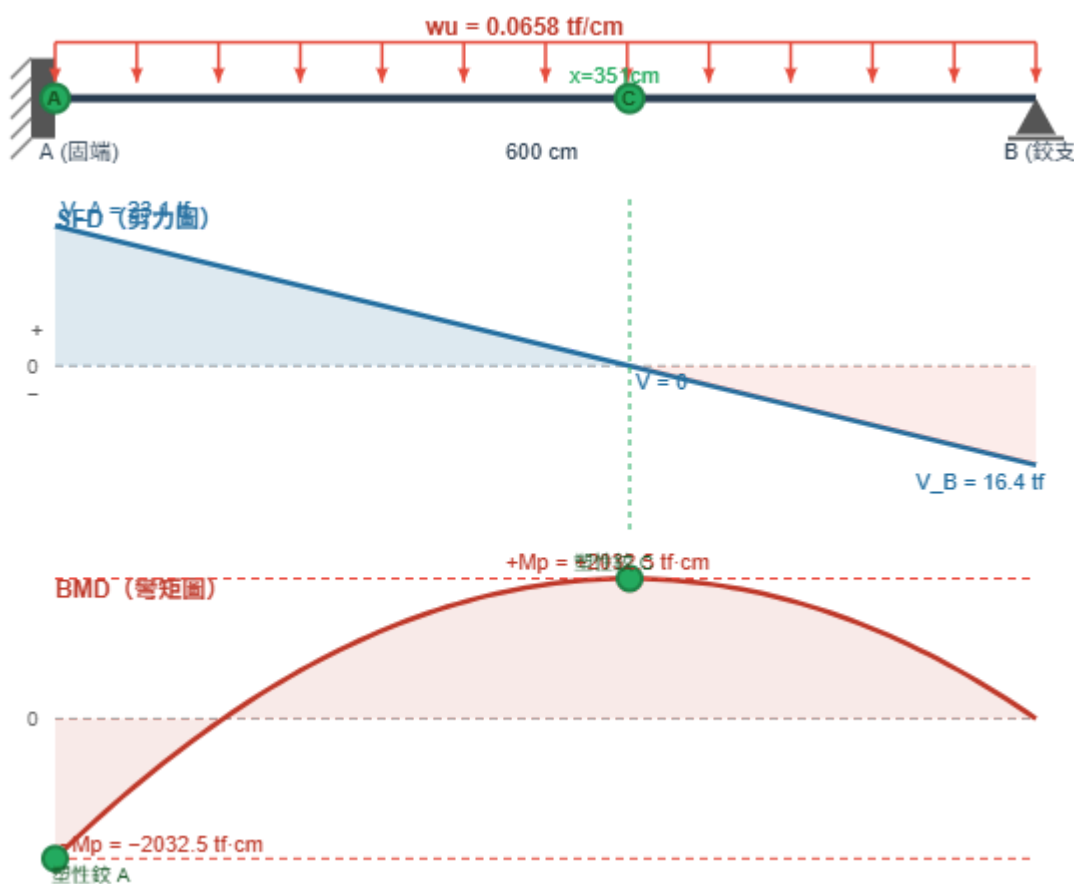
$$\begin{aligned} wu &= (6+4\sqrt{2})M_p/L^2 \\ &= \mathbf{0.0658 \text{ tf/cm}} \\ &= 6.58 \text{ tf/m} \end{aligned}$$

塑性鉸位置

$$\begin{aligned} \text{固端 A: } M_A &= -M_p \\ \text{C 點 } x &= \mathbf{351.5 \text{ cm}} \\ &= L(2-\sqrt{2}) \approx 0.586 L \end{aligned}$$

反力

$$\begin{aligned} R_A &= \mathbf{23.13 \text{ tf}} \\ R_B &= \mathbf{16.33 \text{ tf}} \\ M_A &= \mathbf{-2032.5 \text{ tf}\cdot\text{cm}} \end{aligned}$$



藍色 = SFD (剪力圖) | 紅色 = BMD (彎矩圖) | 綠點 = 塑性鉸 ($M_p = \pm 2032.5 \text{ tf}\cdot\text{cm}$)